6.1 Polynomials Whose Roots Are Roots of Unity

WORKSHEET

The question

STEP III 2010, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q3

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Every nth root of unity satisfies $z^n = 1$. But the smallest positive power that gives 1 might be less than n. For each of the four 4th roots of unity, 1, i, -1, $-\mathrm{i}$, what is that smallest power?

STAGE 1 The first few

Ask yourself Which roots of unity have order 1, 2, 3, 5 and 6? (Draw the 6th roots of unity and sort them by order.)

STAGE 4 No cyclotomic polynomial is a product of two others

Pause and try Suppose $C_q = C_r C_s$. Take a root of C_r. What is its order? What does the identity say about its order? What does that force?

Try this next

1. **Check the factorisation.** Verify that $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = x^6 - 1$, and that $x^{12} - 1$ factorises as in Figure 6.1.1 (in the full section online).
2. **Twice a prime.** Show that if p is an odd prime then $C_{2p}(x) = C_p(-x)$. Check this for $p = 3$.
3. **Totients.** Show that $\varphi(p^k) = p^k - p^{k-1}$ for a prime p, and that $C_{p^k}(x) = C_p(x^{p^{k-1}})$. Use this to find C_9.
4. **A longer one.** HARDERUsing $x^{15} - 1 = C_1 C_3 C_5 C_{15}$, show that $C_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$.

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). The standard undergraduate introduction; cyclotomic polynomials and the regular polygon appear early.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Includes roots of unity and their arithmetic.
- **Marcus du Sautoy, *Finding Moonshine*** (Fourth Estate). A popular account of symmetry and group theory, including the story of Galois.

The full guided solution, and what the question leads to at university, is in Section 6.1:
https://beyondstep.keithmcnulty.org/ch06/cyclotomic.html